Seller Fulfillment Delays and Final Delivery Performance

PBA Final Project - Milestone 3

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Research question and data

How are seller handoff delays associated with final delivery performance, and under what conditions do seller-side delays translate into late customer deliveries?

We use the Olist Brazilian E-Commerce Public Dataset and focus on 86,834 delivered single-item orders. Restricting the sample to single-item orders avoids assigning multiple sellers or shipping deadlines to one order. Seller delay is the difference between the actual carrier handoff timestamp and the seller shipping deadline. Final delivery is classified as late only when the actual delivery calendar date is later than the estimated delivery calendar date. The analysis is descriptive and associational, not causal.

Seller handoff timing

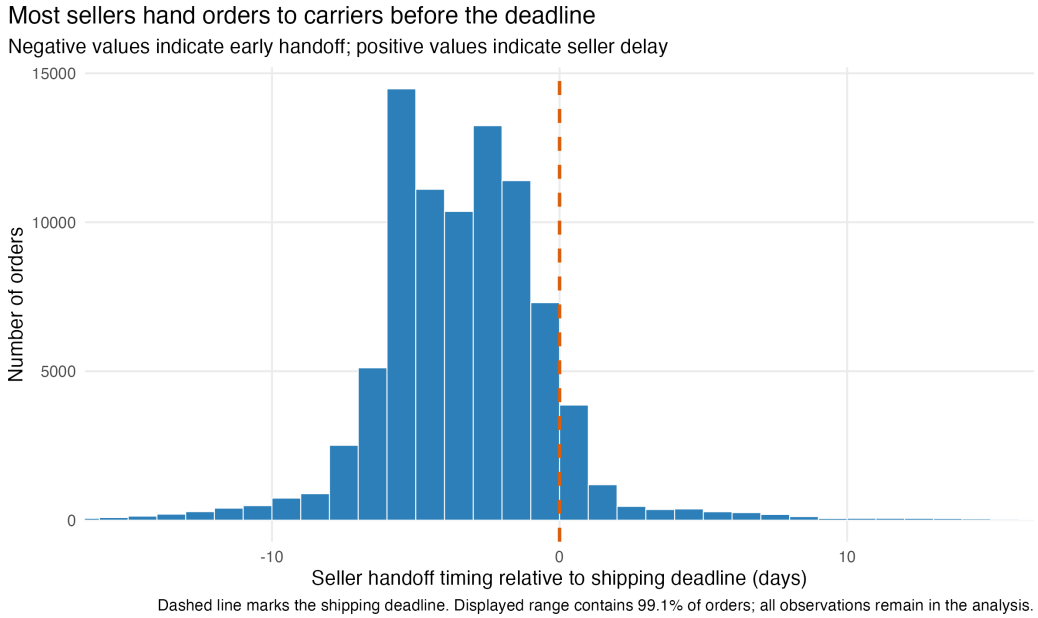


Figure 1: Distribution of seller handoff timing relative to the shipping deadline.

Figure 1 shows that seller handoffs are concentrated before the shipping deadline. The median order was handed off 3.2 days early, while 8.9% were handed off late. The visible right tail motivates examining whether seller delay is associated with final delivery performance.

Final delivery performance by seller status

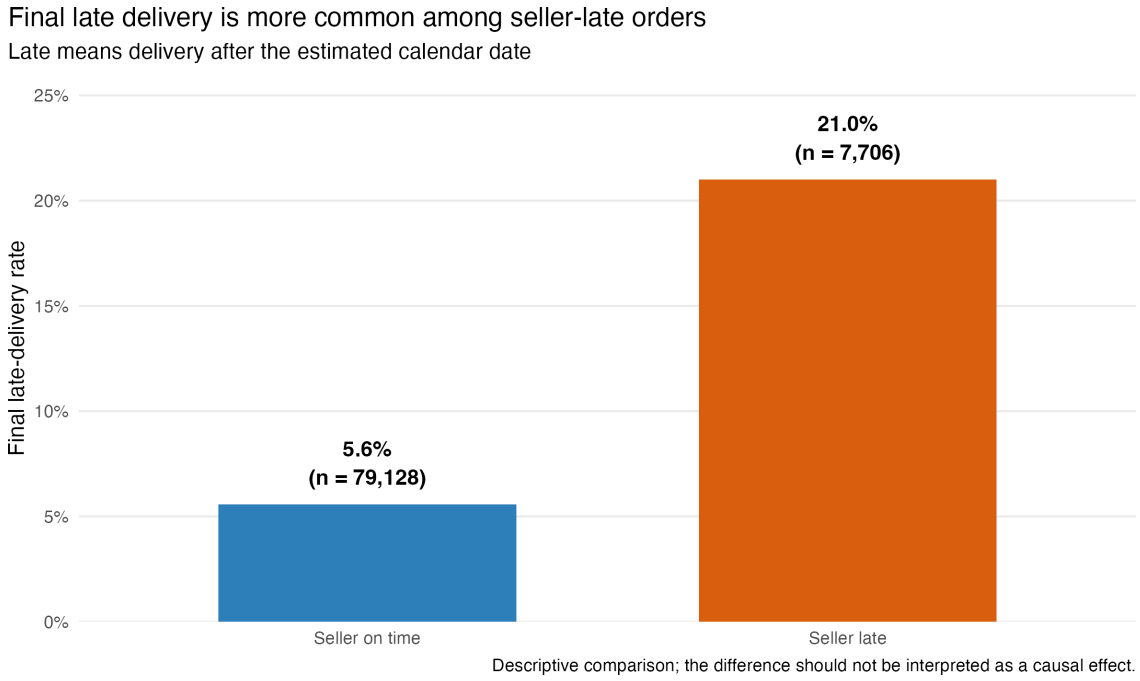


Figure 2: Final late-delivery rate for seller-on-time and seller-late orders.

Figure 2 shows a clear descriptive difference in final delivery performance. The observed final late-delivery rate was 5.6% among seller-on-time orders and 21.0% among seller-late orders, a difference of 15.4 percentage points. This pattern supports further analysis of the association between seller delay and final delivery outcomes. Because the data are observational, the difference does not show that seller delay causes final late delivery.

Planned analysis

The next stage will examine the relationship between `seller_delay_days` and `final_delay_days` using correlation, descriptive group comparisons, and linear regression. We will also examine whether the observed association differs across relevant order and logistics conditions. Any findings will be described as associations rather than causal effects.

Data source

Olist. *Brazilian E-Commerce Public Dataset.* Kaggle. https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce